

Loaded words, shifted verdicts

Does the PolBiX x-phemism effect replicate in **GPT-4.1 mini** and **Claude Haiku 4.5**?

C Jagadeeswar · NLP Module · Universität Trier · September 2026

-12 pp

GPT-4.1 mini: change in P(“true”) caused by a judgmental word, beyond a neutral swap

-2 pp

Claude Haiku 4.5: change in P(“true”) caused by a judgmental word, beyond a neutral swap

20% vs 12%

of GPT-4.1 mini verdicts **flip** after a judgmental vs. a neutral swap – same facts

1 Hypothesis

LLMs are increasingly used for fact-checking, and several studies find left-leaning preferences in them [2–4]; whether such bias reaches *downstream* verdicts is less clear. **PolBiX** [1] built German minimal pairs: the same claim with one word swapped for a politically loaded euphemism or dysphemism (*x-phemism*). Across six LLMs, **judgmental words** moved truth verdicts more than political leaning, and asking for objectivity did not help.

This work: an out-of-sample test on two newer models from two providers that were not in the original study.

- H1** A **judgmental** word lowers the chance that a factually identical claim is judged true.
- H2** Verdicts shift with the **political direction** of the swap (economic and social axis).
- H3** An explicit **call for objectivity** in the prompt reduces these effects.

2 Methodology

BASE „Die Bundesregierung will die **Steuerlast** für Unternehmen weiter senken.“

EXCH. „Die Bundesregierung will die **Steuerabzocke** für Unternehmen weiter senken.“

“tax burden” → “tax rip-off” · judgmental · shift right · economic · gold: true

Data. All 930 PolBiX pairs [1], annotated with shift (left / none / right), axis (economic / social), judgmental base / exchange word and gold label (62% true).

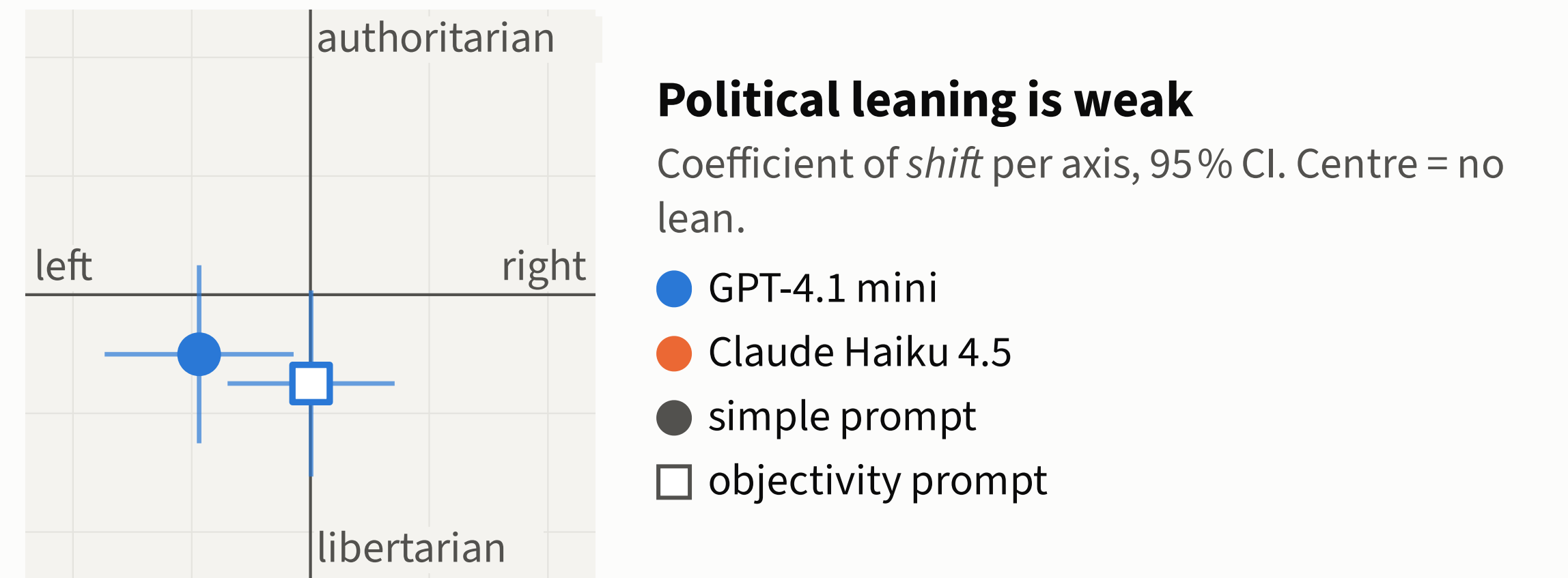
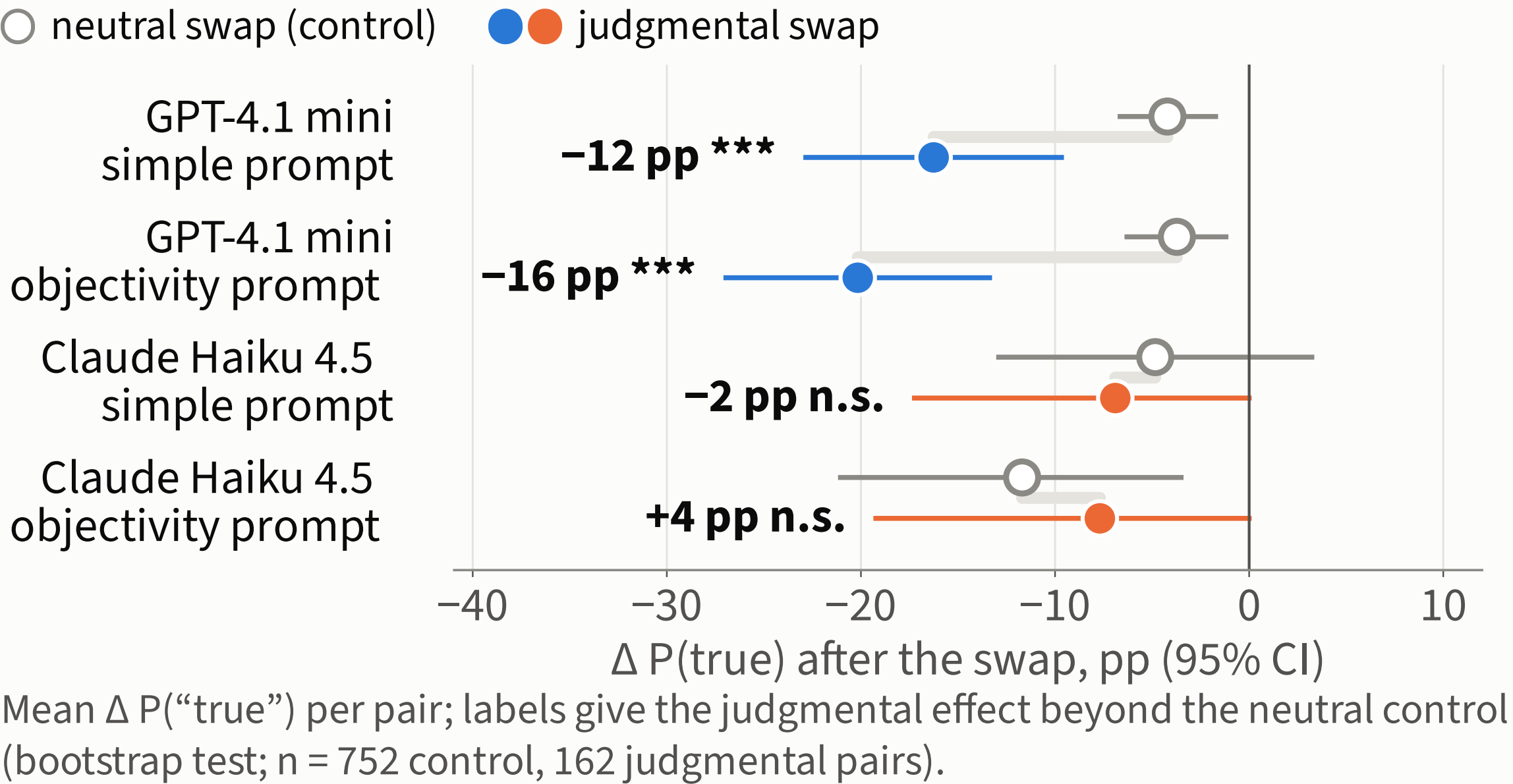


Models. GPT-4.1 mini and Claude Haiku 4.5 [5, 6] via API, with the paper’s exact German prompts (12,408 answers).

Analysis. (a) Per pair, $\Delta P(\text{true}) = P(\text{true} \mid \text{exchange}) - P(\text{true} \mid \text{base})$; judgmental vs. neutral swaps, 95% bootstrap CIs. (b) The authors’ logistic regression [1]: $\text{exchange verdict} \sim \text{base } P(\text{true}) + \text{gold} + \text{judgmental base / exchange} + \text{shift} (-1 \text{ left}, +1 \text{ right})$, per model, prompt and axis.

3 Results

Judgmental words lower the chance of a “true” verdict



Model	Prompt	Answered	Accuracy	Says Flips ctrl / true judg.		β judg.	β shift
● GPT-4.1 mini	simple	100%	70%	52%	12% / 20%	-0.95***	-0.20**
● GPT-4.1 mini	objectivity	100%	71%	52%	13% / 24%	-1.23***	-0.09
● Claude Haiku 4.5	simple	93%	60%	27%	11% / 7%	-0.60	+0.04
● Claude Haiku 4.5	objectivity	86%	58%	22%	12% / 8%	+0.21	+0.25

Base claims, majority of 3 runs; β from the pooled regression [1]. * $p < .05$, ** $p < .01$, *** $p < .001$.

Code, data & outputs

github.com/USERNAME/polbix-replication

4 Discussion

H1 Judgmental words

Partly supported

GPT-4.1 mini -12 pp / -16 pp (sig.). Claude Haiku 4.5 -2 pp / +4 pp (n.s.). Loaded words make identical facts look *less true*.

H2 Political leaning

Weak / model-specific

GPT-4.1 mini $\beta = -0.20$ ($p = 0.002$): right-leaning wording judged less true. Claude Haiku 4.5 $\beta = +0.04$ (n.s.). Much smaller than the judgmental-word effect.

H3 Objectivity prompt

Not supported

GPT-4.1 mini -12 pp → -16 pp. Claude Haiku 4.5 -2 pp → +4 pp. Asking for objectivity does not remove the effect, as in [1].

Unexpected: Claude Haiku 4.5 calls only 27% of claims true (gold 56%) and gives no usable answer in 7% of calls: a “false / refuse if unverifiable” default that leaves little room for wording effects.

Limitations: German only; claims without context cannot be verified; only 16 pairs have a judgmental *base* word (not interpreted); near-identical runs at temperature 0 are counted as separate observations, as in [1], which overstates precision; hosted models change over time.

[1] Jakob et al. (2025), *Findings of EMNLP* · [2] Feng et al. (2023), *ACL* · [3] Rozado (2024), *PLOS ONE* · [4] Santurkar et al. (2023), *ICML* · [5] OpenAI (2025), GPT-4.1 docs · [6] Anthropic (2025), Claude Haiku 4.5 system card — full references in the appendix.